

The Fundamentals of RDMA Programming

Computers in a network can exchange data in the main memory with the involvement of the processor, cache or OS by using a technology called remote direct memory access (RDMA). This frees up resources, improving the throughput and performance while facilitating faster data transfer.



Remote direct memory access (RDMA) technology increases the speed of server-to-server data movement through better utilisation of network infrastructure without CPU intervention. The network adapter transfers data directly to or from the application memory without interrupting other parallel operations of the system. RDMA technology is widely used in enterprise data centres and high performance computers (HPC) because of its high-throughput and low-latency networking.

This article will enable app developers to start programming RDMA apps even without any experience with it. Before we start, let's have a brief introduction to InfiniBand (IB) fabrics — its features and components.

InfiniBand (IB)

InfiniBand is an open industry-standard specification for data flow between server I/O and inter-server communication. IB supports RDMA and offers high-speed, low latency, low CPU overhead, high efficiency and scalability. The transfer speed of InfiniBand ranges from 10Gbps (SDR) to 56Gbps (FDR) per port.

Components of InfiniBand

Host channel adapter (HCA): This provides an address

translation mechanism under the control of the operating system, which allows an application to access the HCA directly. The same address translation mechanism is the means by which an HCA accesses memory on behalf of a user level application. The application refers to virtual addresses, while the HCA has the ability to translate these addresses into physical addresses in order to effect the actual message transfer.

Switches: IB switches are conceptually similar to standard networking switches but are designed to meet IB performance requirements. They implement the flow control of the IB Link Layer to prevent packet dropping and to avoid congestion. They also have adaptive routing capabilities and advanced quality of service. Many switches include a subnet manager, at least one of which is required to configure an IB fabric.

Range extenders: InfiniBand range extension is accomplished by encapsulating the InfiniBand traffic onto the WAN link and extending sufficient buffer credits to ensure full bandwidth across the WAN.

Subnet managers: The IB subnet manager is based on the concept of software defined networking (SDN), which eliminates interconnect complexity and enables the creation of very large scale compute and storage infrastructures. The IB subnet manager assigns local identifiers (LIDs) to each port connected to the InfiniBand fabric, and develops a